

Homework 10

1. Let U be the positive octant of $\mathbb{R}^3$. Let D be the distribution spanned by the vector fields

$$X = y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y}, \quad Y = z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z}.$$

(a.) We need to show D is involutive. It suffices to verify that the Lie bracket $[X, Y]$ is contained in D , since any element of D is a linear combination of X and Y , and by the bilinearity of the Lie bracket. We calculate that

$$[X, Y] = y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y} = \frac{x}{z} X + \frac{y}{z} Y.$$

These coefficients are smooth functions on the open set U .

(b.) Notice both X and Y are perpendicular to the vector field $A = x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z}$, since for example, at $p \in U$,

$$X_p \cdot A_p = 0.$$

Then D_p is orthogonal to the vector p . But D_p is two dimensional, so it is the entire orthogonal complement of p . Then any integral submanifold of D must have its tangent bundle orthogonal to the radial vector field A . This is true for a sphere of any radius, restricted to U .

Moreover, any submanifold which does not lie in a sphere must have varying radius. Suppose M is such a submanifold of U , and choose points p and q with different distances from the origin. Consider a path from p to q lying in the plane defined by the points p , q , and the origin. After converting to ordinary 2-d polar coordinates on this plane, by the mean value theorem, the velocity vector of this path must have non-zero radial component for some t , say at a point p' . But this velocity vector must lie in the tangent space of M , and thus M is not an integral manifold.

Then the only integral submanifolds of D are subsets of spheres. A maximal integral submanifold is thus an entire octant of a sphere of nonzero radius.

2. Let D be the distribution of $\mathbb{R}^3$ spanned by

$$X = \frac{\partial}{\partial x} + yz \frac{\partial}{\partial z} \quad Y = \frac{\partial}{\partial y}.$$

(a.) An integral submanifold of D passing through the origin is the plane $z = 0$. The z component of X vanishes when $z = 0$, and its tangent space is clearly spanned by $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$.

(b.) We can compute the bracket

$$[X, Y] = z \frac{\partial}{\partial z},$$

which does not lie in the span of X and Y , as there is no way to eliminate either the x or the y components by taking linear combinations. This demonstrates that a distribution may have integral submanifolds at some locations, but still not be involutive.

3. Let I be an ideal of forms on M locally generated by $\{\omega^i\}_1^r$ on an open set U .

First suppose I is a differential ideal. Let $\omega := \omega^1 \wedge \cdots \wedge \omega^r$. Using the product rule for forms, we can iterate

$$\begin{aligned} d\omega &= d(\omega^1 \wedge \cdots \wedge \omega^r) = d\omega^1 \wedge (\omega^2 \wedge \cdots \wedge \omega^r) + \omega^1 \wedge d(\omega^2 \wedge \cdots \wedge \omega^r) = \\ &\quad \vdots \\ &= d\omega^1 \wedge \omega^2 \wedge \cdots \wedge \omega^r + \omega^1 \wedge d\omega^2 \wedge \cdots \wedge \omega^r + \cdots + \omega^1 \wedge \cdots \wedge d\omega^r. \end{aligned}$$

Then using the fact that I is a differential ideal, write $d\omega^i = \sum \alpha_j^i \wedge \omega^j$ for each i . Then many of the terms above vanish, leaving only

$$\sum \alpha_i^i \wedge \omega = \alpha \wedge \omega$$

where $\alpha = \sum \alpha_i^i$.

Conversely suppose $d\omega = \alpha \wedge \omega$ for some α . We can complete the forms to a basis $\{\omega^k\}_1^n$ by restricting U . For each i write $d\omega^i = \sum_I a_I^i \omega^I$, where I ranges over all 2 element subsets of $[1, \dots, n]$. Then

$$d\omega = \sum_i (-1)^{i-1} d\omega^i \wedge (\omega^1 \wedge \dots \wedge \widehat{\omega^i} \wedge \dots \wedge \omega^n) = \sum_{I,i} a_I^i \omega^I \wedge (\omega^1 \wedge \dots \wedge \widehat{\omega^i} \wedge \dots \wedge \omega^n). \quad (1)$$

Also write $\alpha = \sum_1^n \alpha_i \omega^i$. Then we have

$$d\omega = \sum_1^n \alpha_i \omega^i \wedge \omega = \sum_{r+1}^n \alpha_i \omega^i \wedge \omega. \quad (2)$$

Then the sums 1 and 2 are equal. Both are composed of basis elements of the $r+1$ forms induced by the ω^i 's, so by linear independence, each a_I^i for which both elements of I are greater than r must be zero. Then every term in the expressions for the $d\omega^i$ contains an element of the generating set $\{\omega^1, \dots, \omega^r\}$, and can be written in the form

$$d\omega^i = \sum_1^r \beta_i \wedge \omega^i$$

for some β_i . This proves I is differential for the generating forms ω^i . For arbitrary elements of I , we can prove their differential is in I by writing them as combinations of wedges with basis elements and expanding using the definition of the differential.

4. Let D be a smooth distribution of rank k on M an n dimensional manifold. Let $\omega^1, \dots, \omega^{n-k}$ be its defining forms within a set U .

First suppose D is involutive. The set of forms which vanish on D is an ideal I_D . It is generated by the defining forms $\omega^1, \dots, \omega^{n-k}$. From the notes, we know that D is involutive if and only if I_D is differential. Then we can write

$$d\omega^i = \sum_{j=1}^{n-k} \alpha_j^i \wedge \omega^j$$

for some α_j^i . Then it follows that

$$d\omega^i \wedge \omega^1 \wedge \dots \wedge \omega^{n-k} = 0$$

after expanding $d\omega^i$, since all terms contain duplicate indices.

Conversely suppose $d\omega^i \wedge \omega^1 \wedge \dots \wedge \omega^{n-k} = 0$ for all $i = 1, \dots, n-k$. It suffices to prove $d\omega^i \in I_D$ for all i by a theorem in Lee. If we expand $d\omega^i$ after completing our set to a basis, there cannot be any terms which do not contain some ω^i for $i \in [1, \dots, n-k]$, otherwise $d\omega^i \wedge \omega^1 \wedge \dots \wedge \omega^{n-k}$ would not vanish.

6. Let D be a distribution on M and X a vector field, and e^{tX} the local flow of X . (a.) suppose $(e^{tX})_* D = D$ for small t . The Lie derivative is defined as

$$\mathcal{L}_X(Y) = \frac{d}{dt}(e^{tX})_* Y|_{t=0} = [X, Y].$$

The condition above guarantees that the time derivative will also be in D , by writing as a difference quotient. So $[X, Y] \in D$ for all $Y \in D$.

(b.) Suppose $[X, Y] \in D$ when $Y \in D$. If e^{tX} is defined, for $Y \in D$, we have

$$[X, Y] = \frac{d}{dt}(e^{tX})_*Y|_{t=0}.$$

Let $Y_t = (e^{tX})_*Y$. However by the property of local flows we can take the restriction at any time close to zero. Then the problem is to find a flow for which $[X, Y]$ is the infinitesimal generator. But this can be solved locally within D since $[X, Y] \in D$. Thus D is invariant under pullback of Y by the flow of X .

(c.) That $[X, Y]$ is an infinitesimal symmetry of D when X and Y are follows from writing

$$[Z, [X, Y]] = -[X, [Y, Z]] - [Y, [Z, X]],$$

where $Z \in D$, and noting first that the inner brackets are in D , then the outer ones.